

Basic Principles of Medicine II

Module: Nervous System and Behavioral Sciences | Lecture NB 12

Face and Scalp

Dr. Michael Montalbano

mmontall@sgu.edu

Department of Anatomical Sciences

School of Medicine, St. George's University

Copyright

All year-1 course materials, whether in print or online, are protected by copyright. The work, or parts of it, may not be copied, distributed, or published in any form, printed, electronic, or otherwise.

As an exception, students enrolled in year 1 of St. George's University School of Medicine and their faculty are permitted to make electronic or print copies of all downloadable files for personal and classroom use only, provided that no alterations to the documents are made and that the copyright statement is maintained in all copies.

'View only' files, such as lecture recordings, are explicitly excluded from download, and creating copies of these recordings by students and other users is strictly illegal.

The author of this document has made the best effort to observe current copyright law and the copyright policy of St. George's University. Users of this document identifying potential violations of these regulations are asked to bring their concerns to the attention of the author.

Recommended reading

Drake RL, Vogl AW, Mitchell AWM. Gray's anatomy for students. 4th ed. Elsevier. Chapter 8.

Siegel and Sapru Essential Neuroscience 4th ed. Chapter 13.

Neuroanatomy in clinical context. An atlas of structures, sections, systems and syndromes. Haines DE. 10th ed. Wolters Kluwer. Chapter 3

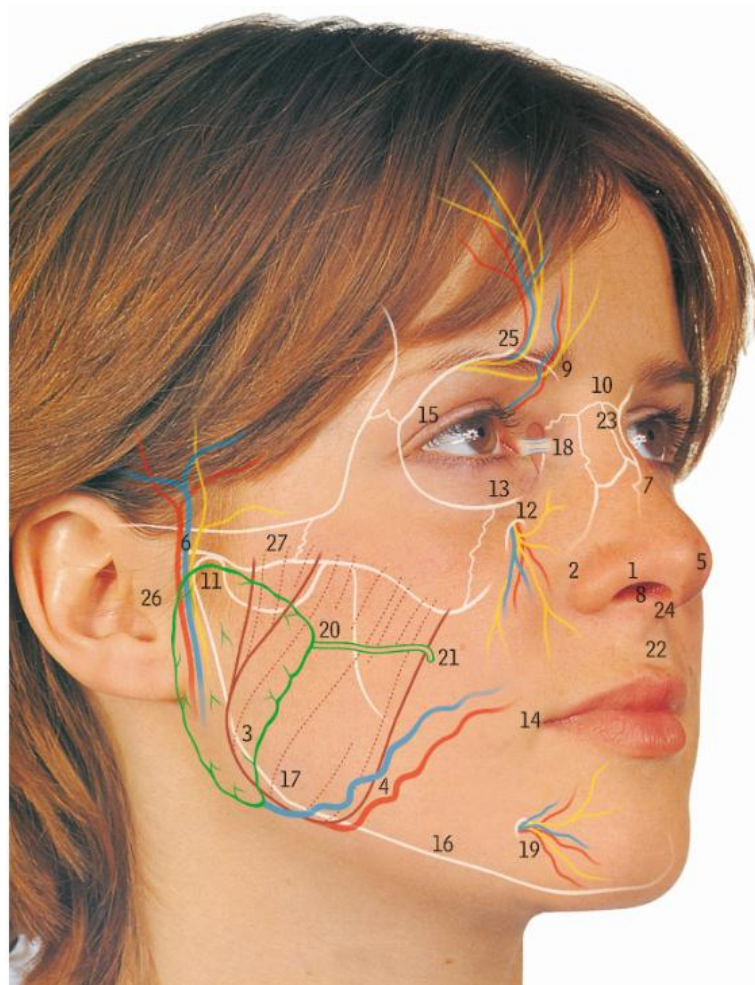


Pay special attention to the green 'In the Clinic' boxes which contain the clinical correlates for this topic.

Objectives

SOM.MKII.BPM2.1.NB.1.ANAT.0082	List the locations, innervations and actions of the major muscle groups of facial expression
SOM.MKII.BPM2.1.NB.1.ANAT.0083	Describe the course of the facial artery and vein
SOM.MKII.BPM2.1.NB.1.ANAT.0084	Describe the location, innervation and blood supply of the parotid gland
SOM.MKII.BPM2.1.NB.1.ANAT.0085	Describe the exit of the facial nerve from the temporal bone via the stylomastoid foramen and its relationship to structures as it courses towards and through the parotid gland
SOM.MKII.BPM2.1.NB.1.ANAT.0086	Describe the extracranial course of the facial nerve and its distribution
SOM.MKII.BPM2.1.NB.1.ANAT.0087	Describe the clinical significance of the buccal fat pad
SOM.MKII.BPM2.1.NB.1.ANAT.0088	Describe the course of the parotid duct and its opening in the oral vestibule
SOM.MKII.BPM2.1.NB.1.ANAT.0089	Discuss how pathology of the parotid gland may lead to facial palsy
SOM.MKII.BPM2.1.NB.1.ANAT.0090	Describe the five layers of the scalp, its innervation, blood supply and venous return
SOM.MKII.BPM2.1.NB.1.ANAT.0091	Describe the “danger zone” of the scalp and its clinical significance
SOM.MKII.BPM2.1.NB.1.ANAT.0092	Describe how infections of the scalp and the face may spread into the cranial cavity
SOM.MKII.BPM2.1.NB.1.ANAT.0093	Describe the lymphatic drainage of the superficial and deep structures of the face and scalp
SOM.MKII.BPM2.1.NB.1.ANAT.0094	Describe the causes and locations of caput succedaneum, cephalohematoma and subgaleal hematoma
SOM.MKII.BPM2.1.NB.1.ANAT.0095	Describe the sensory innervation of the face
SOM.MKII.BPM2.1.NB.1.ANAT.0227	Identify major anatomical structures of the face and scalp in MRI, CT and X-Ray imaging
SOM.MKII.BPM2.1.NB.1.ANAT.0225	Identify the spaces of the skull on X-ray, CT and MRI modalities

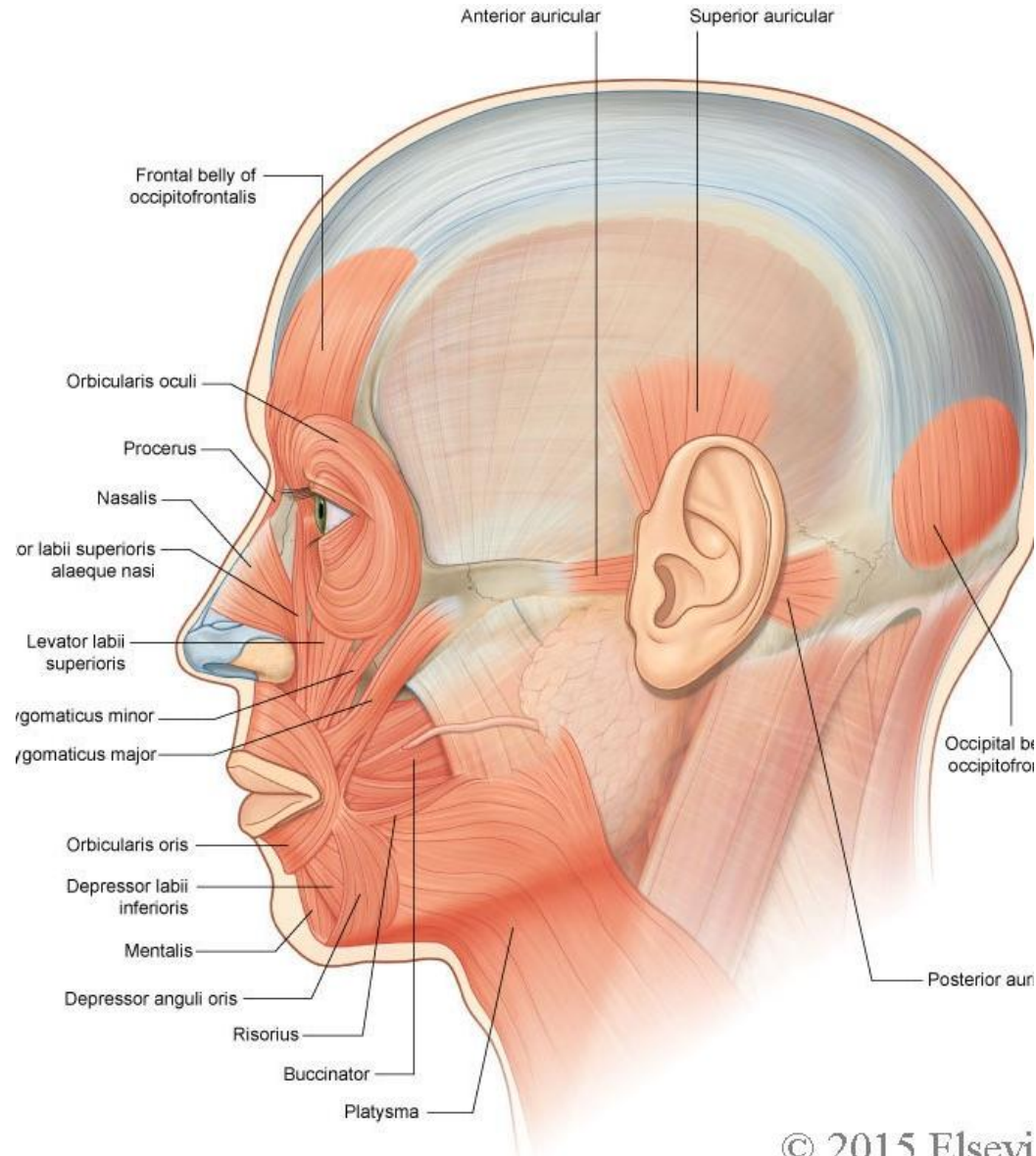
Overview



- Face
 - Muscles and parotid gland
 - Vasculature
 - Innervation
- Scalp
 - Layers
 - Vasculature
 - Innervation
- CN: V, VII

Muscles of the Face - Overview

- Four Groups
 - I. Orbital
 - II. Nasal
 - III. Oral
 - IV. Other
- Develop from
2nd pharyngeal arch
- Action
 - muscles of facial expression
 - sphincters around eyes and mouth



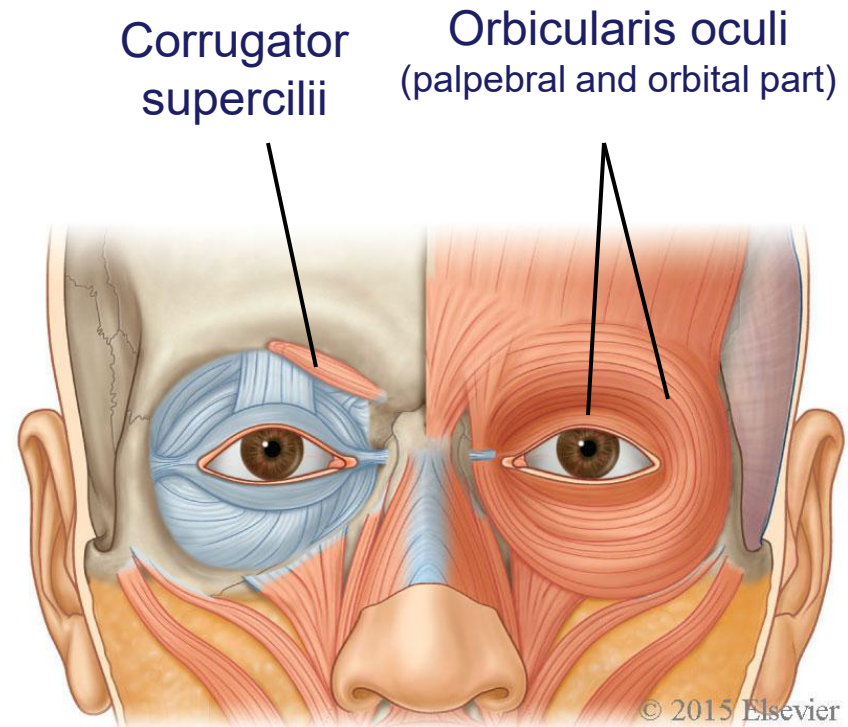
I. Orbital Group of Facial Muscles

- **Orbicularis oculi**

- completely surrounds each orbit
- orbital part
 - encircles the orbital opening and beyond the rim
 - closes the eye forcefully
- palpebral part
 - located in the eyelids from the medial corner of eye across each lid to attach laterally
 - closes the eye gently

- **Corrugator supercilii**

- deep to the eyebrows
- draws the eyebrow down and medial, wrinkles the skin above the nose during frowning



II. Nasal Group of Facial Muscles

Nasalis

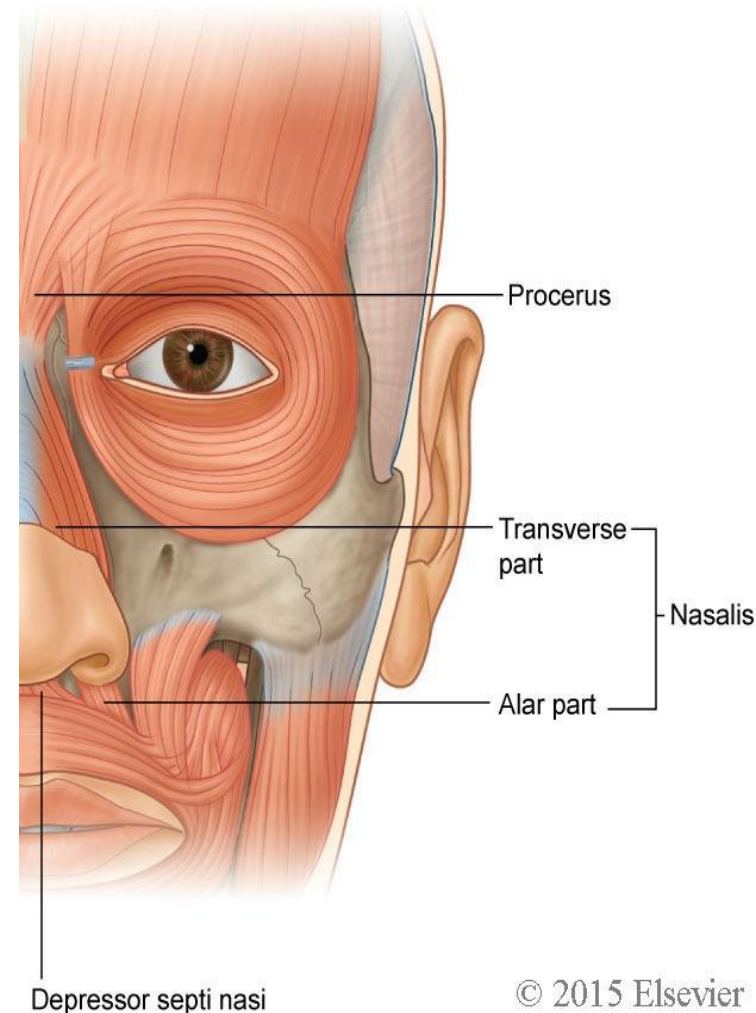
- Transverse part: compresses the nares
- Alar part: draws the alar cartilages down and lateral, and opens the nares

Procerus

- Arises from the nasal bone
- Inserts into the skin of the forehead, between the eyebrows
- Draws the medial border downward, wrinkling the skin superficial to the nasal bone
- Active during frowning

Depressor septi nasi

- Arises from maxilla and inserts into nasal septum
- Widens the nares by pulling the nose inferiorly



III. Oral Group of Facial Muscles

LOWER GROUP:

Depressor anguli oris - depresses corner of mouth (frowning)

Depressor labii inferioris - depresses lower lip laterally

Mentalis - positions lip when drinking from a cup, or pouting

UPPER GROUP:

Risorius – grimaces, wide smiles

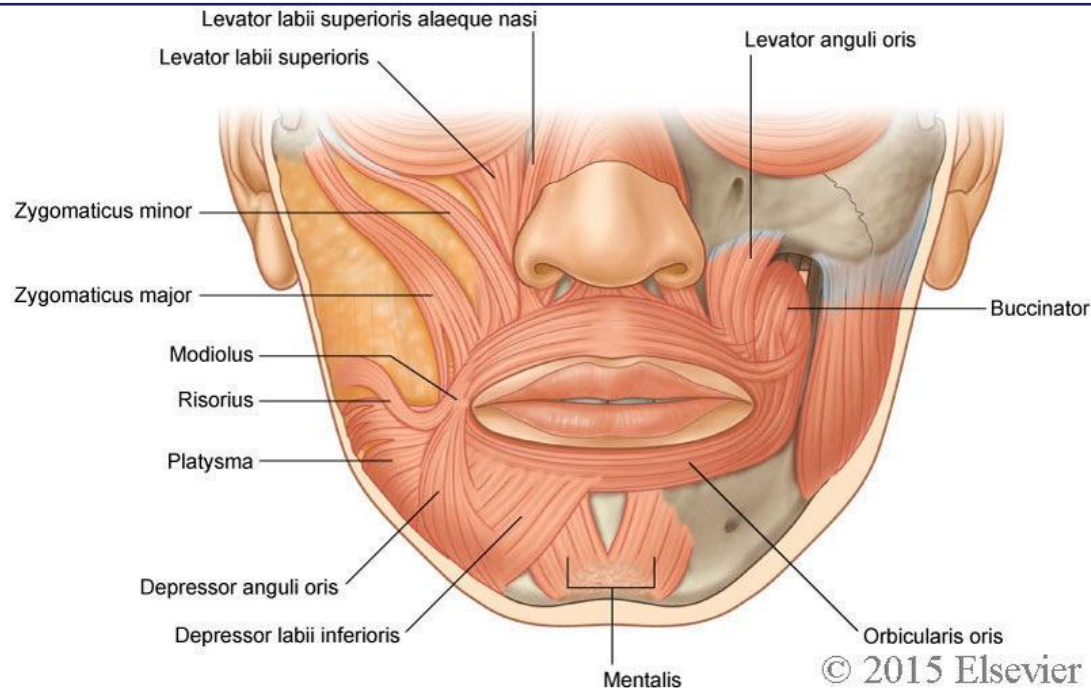
Zygomaticus major/minor – smiling, laughing

Levator labii superioris - deepens furrow between nose and mouth

Levator labii superioris alaeque nasi - flaring the nostrils

Levator anguli oris - elevates corner of mouth

Orbicularis oris - narrows the mouth and closes the lips (pursed lips)



© 2015 Elsevier

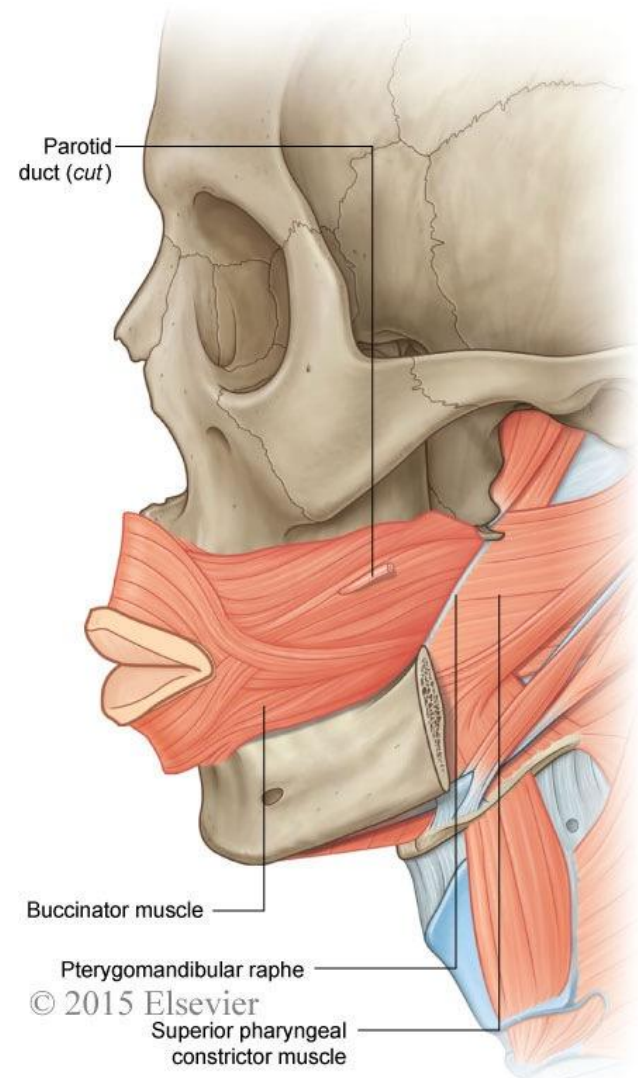
Copyright © 2015, 2010, 2005 by Churchill Livingstone, an imprint of Elsevier Inc.

III. Oral Group of Facial Muscles

Buccinator

- Location
 - muscular component of the cheek
 - occupies space between mandible and maxilla deep to other muscles of the face
- Attachments
 - posterior maxilla & mandible opposite molar teeth
 - pterygomandibular raphe
 - blends in with fibers of the orbicularis oris
 - upper fibers enter the lower lip
 - lower fibers enter the upper lip
- Actions
 - presses cheeks against teeth (one of the muscles of mastication)
 - forceful expulsion of air from cheeks
 - role in mastication

Parotid duct pierces the buccinator muscle



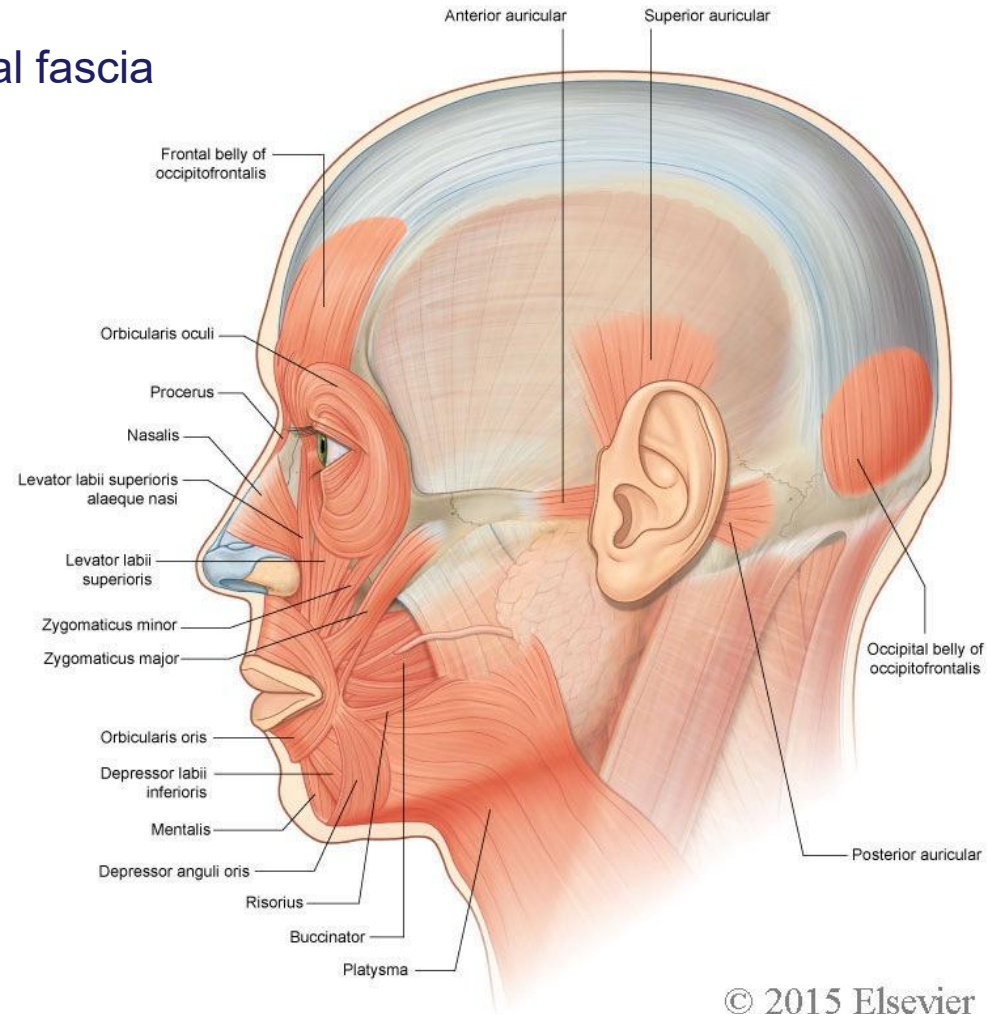
© 2015 Elsevier

Superior pharyngeal
constrictor muscle

Copyright © 2015, 2010, 2005 by Churchill Livingstone, an imprint of Elsevier Inc.

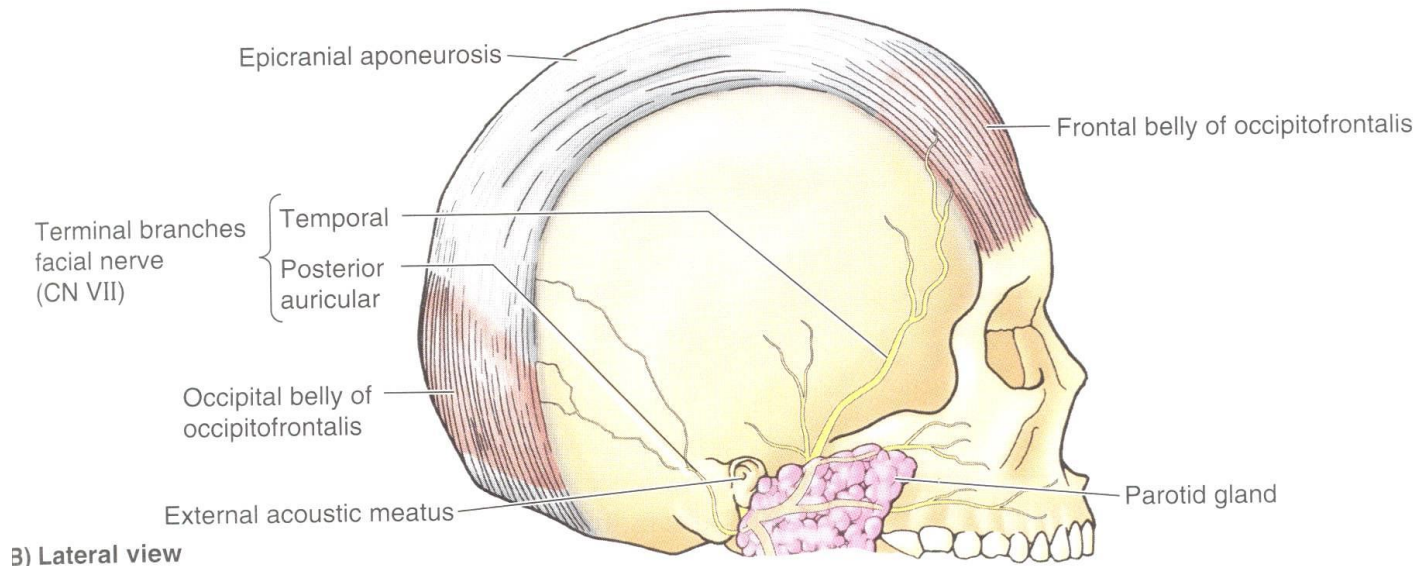
IV. Other Associated Muscles -

- Not facial, but derived from 2nd pharyngeal arch and share same innervation
- Platysma
 - Imbedded in superficial cervical fascia
- Auricular muscles
 - Anterior auricular
 - Superior auricular
 - Posterior auricular
- Posterior belly of digastric
- Stylohyoid
- Occipitofrontalis
 - Frontal belly
 - Occipital belly



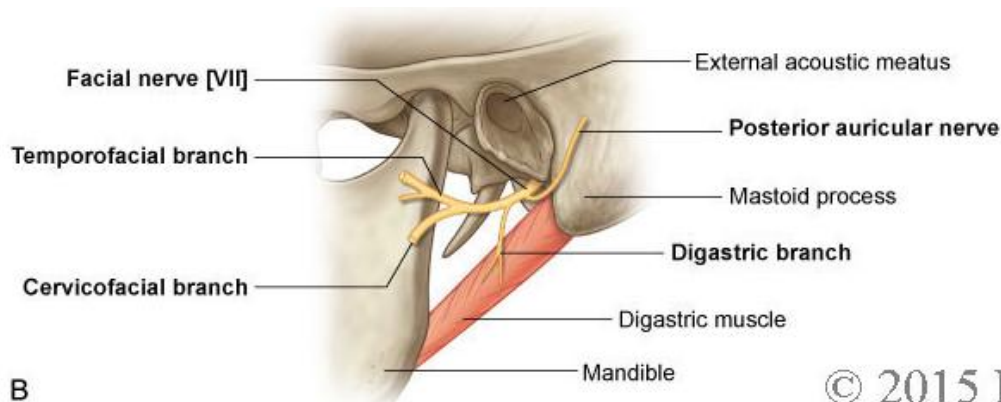
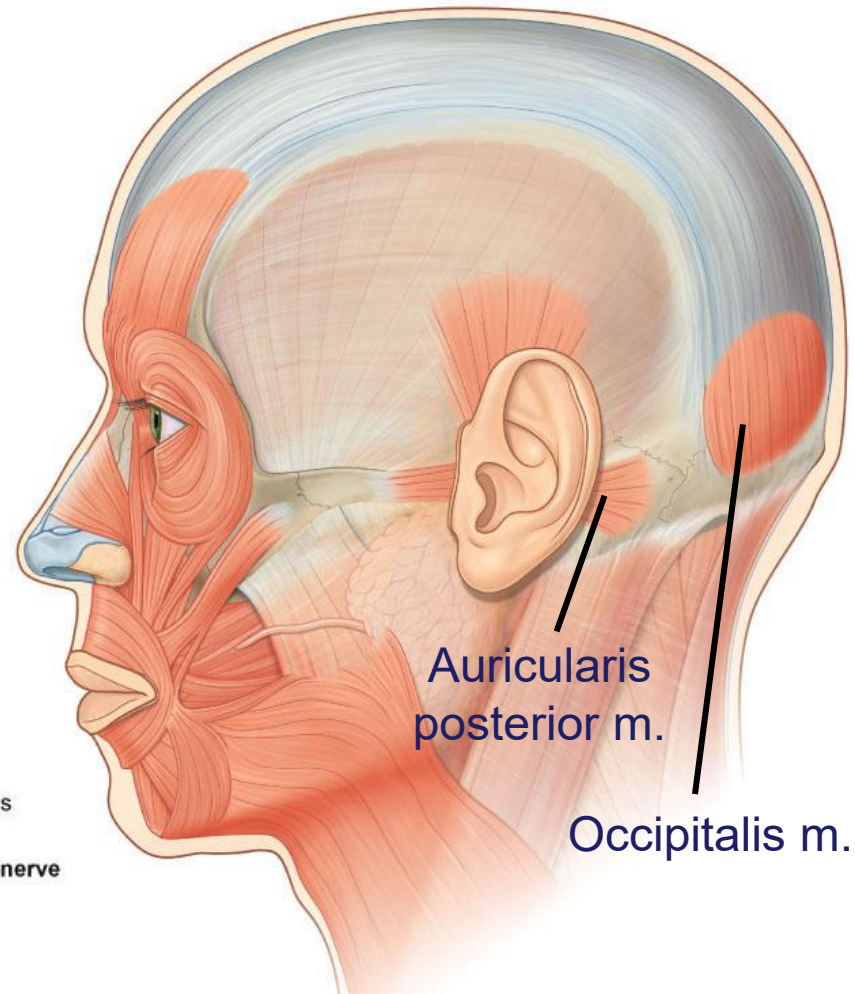
Occipitofrontalis

- Occipitalis
 - Attached to superior nuchal line and mastoid process of the temporal bone to epicranial aponeurosis
 - Innervated by posterior auricular branch of VII
- Frontalis
 - Attached to skin of eyebrows to epicranial aponeurosis
 - Innervated by temporal branch of VII
- Separated by an aponeurosis
 - Epicranial aponeurosis (galea aponeurotica) – embedded in scalp
- Action
 - Moves the scalp, wrinkles the forehead, and raises the eyebrows



Motor Innervation – VII

- Exits the skull via stylomastoid foramen
- Initial extracranial branches
 - **Posterior auricular nerve**: to auricularis posterior and occipitalis
 - Branch to **posterior belly of digastric**
 - Branch to **stylohyoid**
- Courses anteriorly and enters the parotid gland
- The rest of VII innervates the muscles of facial expression via 5 branches

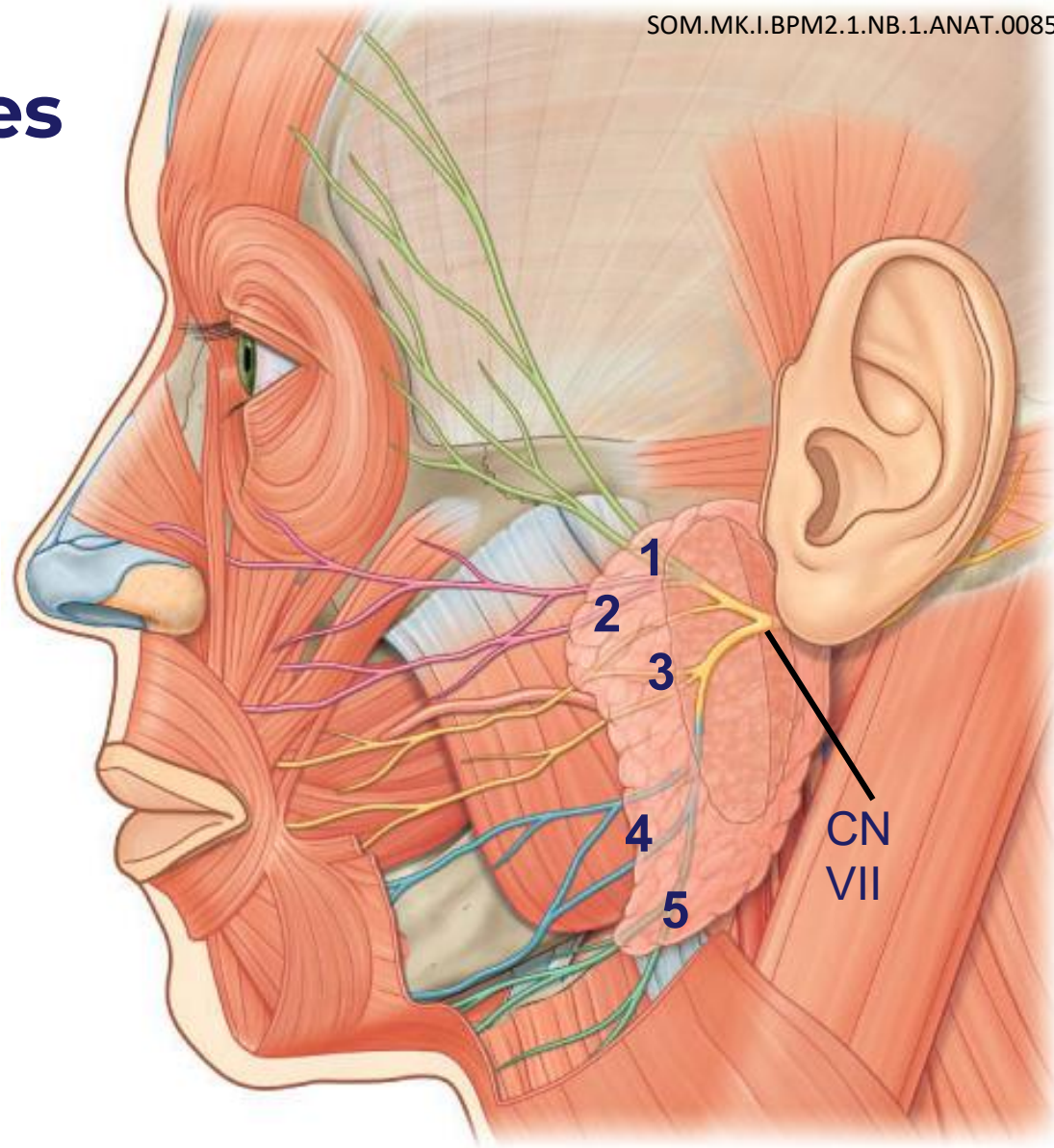


5 motor branches

CN VII enters the parotid gland and divides into :

1. Temporal
2. Zygomatic
3. Buccal branch
4. Marginal mandibular
5. Cervical

“Two Zebras Bit My Clavicle”



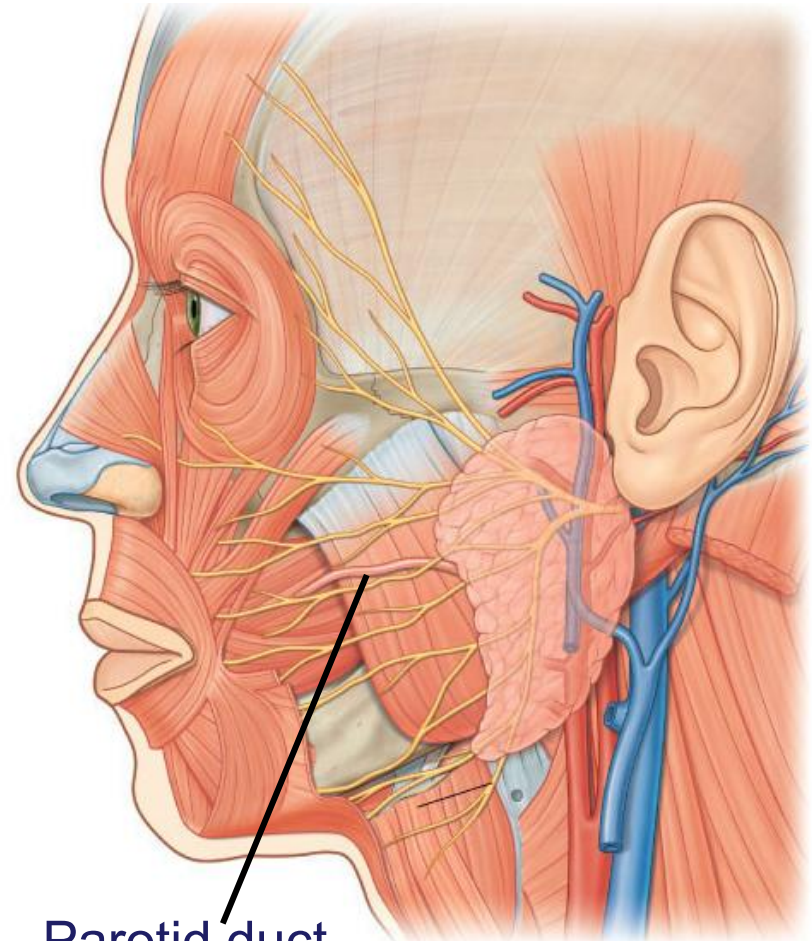
Parotid Gland

- Anterior to lower half of ear
- Extends from zygomatic arch down to lower border of mandible
- Covers portion of SCM and masseter

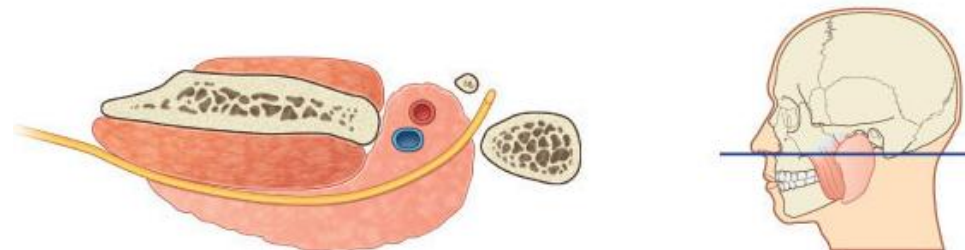
Parotid (Stensen's) duct:

Pathway to oral cavity:

- Leaves anterior edge of gland midway between zygomatic arch and corner of mouth
- Crosses the anteromedial limit of masseter,
- Penetrates the buccal fat pad and buccinator
- Opens into oral cavity opposite upper second molar tooth



Parotid duct

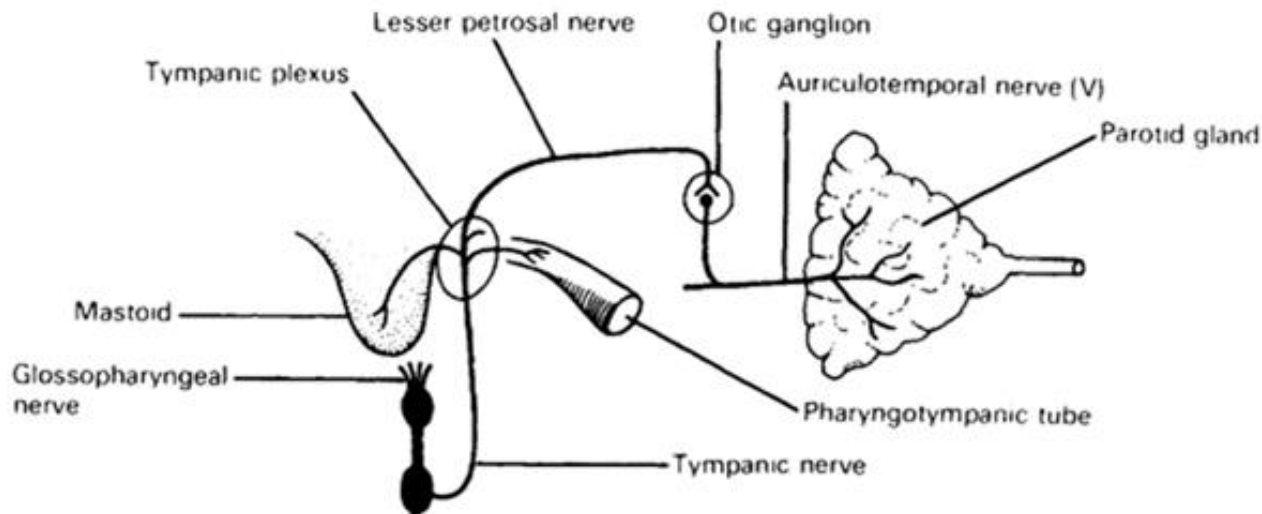


Parotid Gland

Innervation & Blood Supply

Innervation

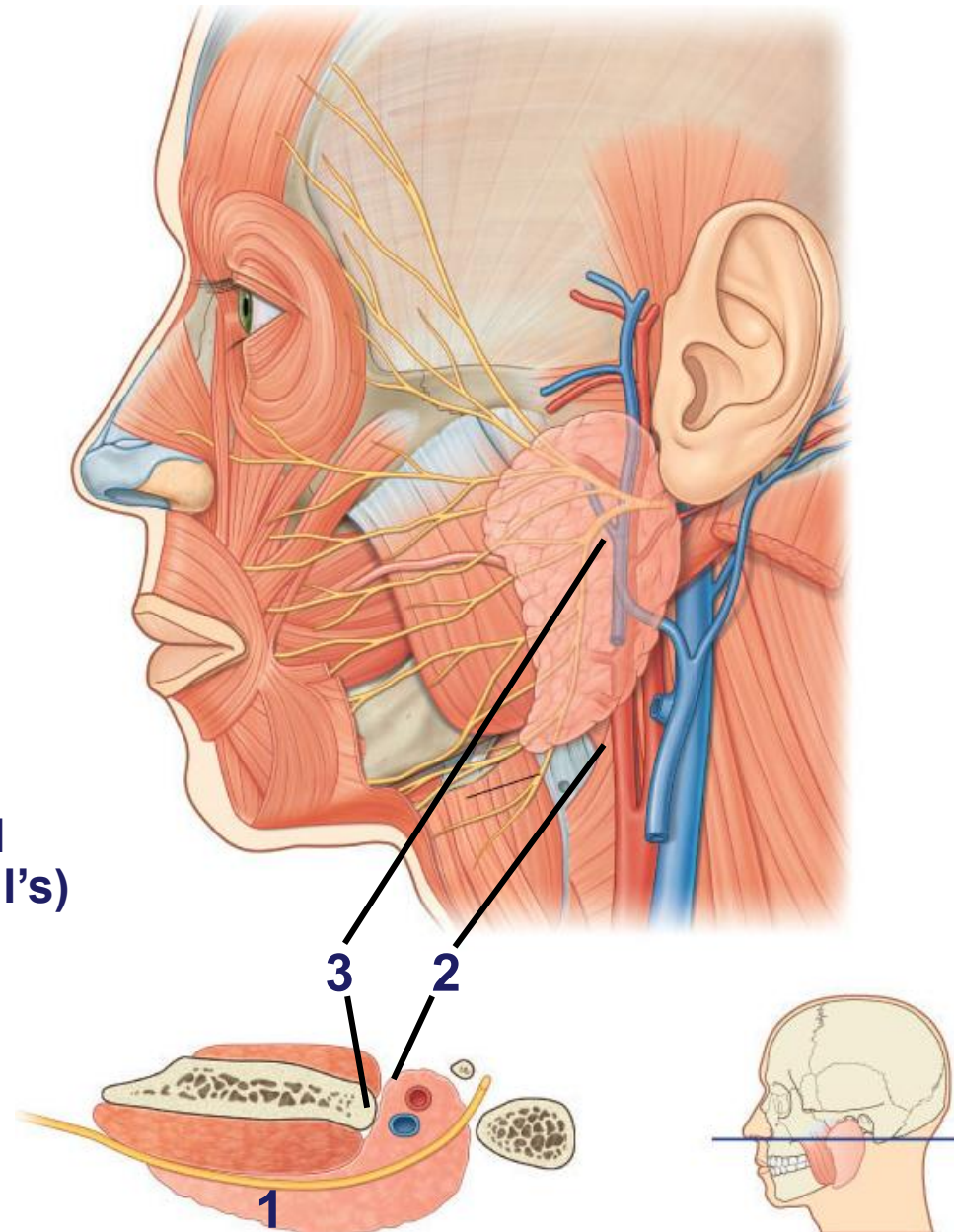
- Sensory: auriculotemporal nerve (V_3)
- Secretomotor: auriculotemporal nerve
 - Preganglionic parasympathetic fibers from IX
 - Postganglionic parasympathetic fibers synapse at the **otic ganglion**
- Blood supply
 - through numerous arteries that pass through the gland



Parotid Gland

Structures passing through the gland:

1. External Carotid Artery and its branches
 - Superficial temporal artery
 - Posterior auricular
 - Transverse facial
 - Maxillary
2. Retromandibular vein
 - Superficial temporal vein
 - Maxillary vein
3. Facial nerve (CN VII)
 - Pathology or trauma of parotid gland may result in **facial (Bell's) palsy**



Clinical Correlates - Parotid Gland

Parotid gland tumors:

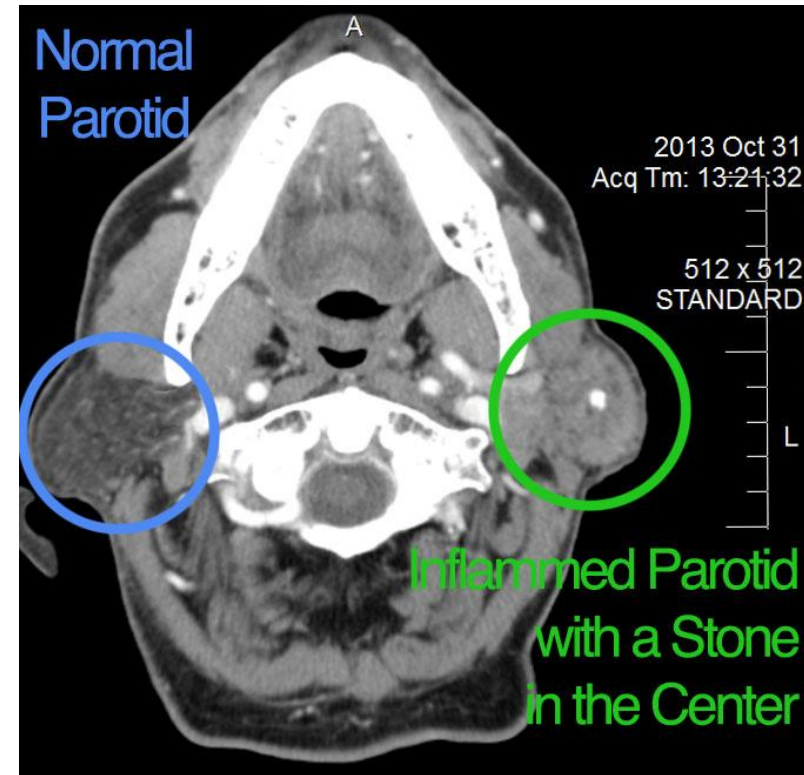
- typically involve the superficial gland

Parotid gland stones (sialolithiasis):

- typically within the main parotid duct.
- involves intense pain
 - reproducible by sialalogue stimulation testing, i.e. squirting lemon juice in the the patient's mouth.
- surgical excision necessary

Parotitis:

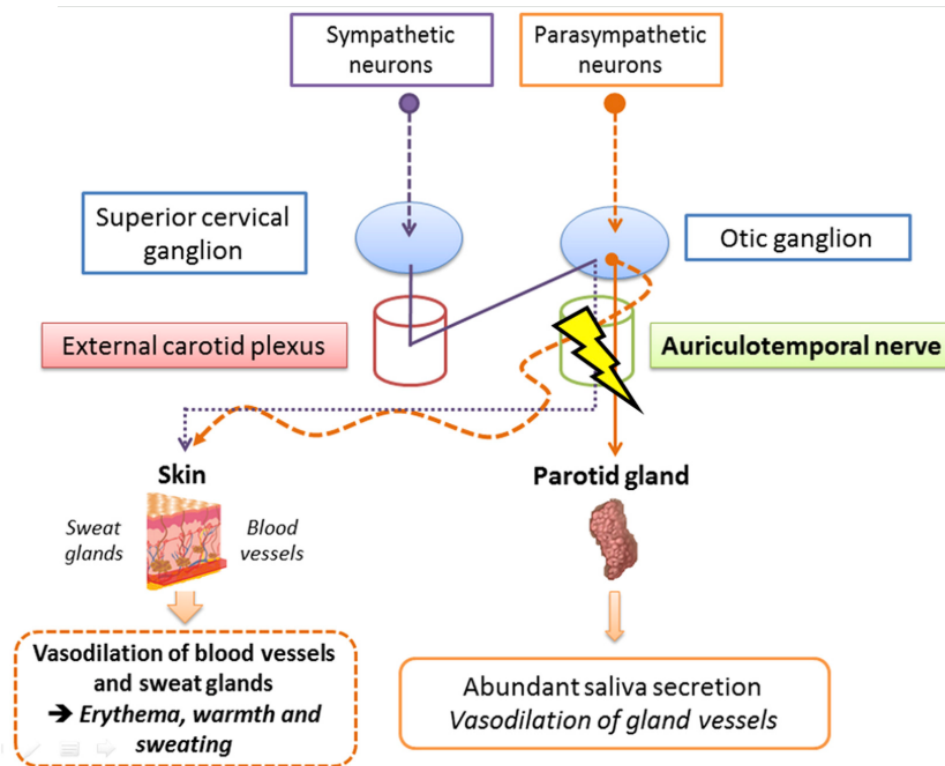
- causes include parotid duct obstruction or inflammatory conditions
- results in pain, glandular enlargement and swelling, dry mouth, and sometimes fever.



<https://www.parotidsurgerymd.com/diagnosis>

Clinical Correlate – Auriculotemporal (Frey's) syndrome

- Usually occurs after parotidectomy (surgical removal of parotid gland)
- After the procedure, axons of the postganglionic parasympathetic fibers from the auriculotemporal nerve regrow into the superficial fascia
 - Can lead to formation of synapses with sympathetic cholinergic sweat glands
- Presents with **“gustatory sweating”**: sweating of the skin of the face in response to the smell.



Trauma of sympathetic postganglionic neurons



Aberrant parasympathetic postganglionic neurons

From DOI: 10.1016/j.jpeds.2016.03.070

Sensory distribution of CN V

Ophthalmic nerve (CN V₁)

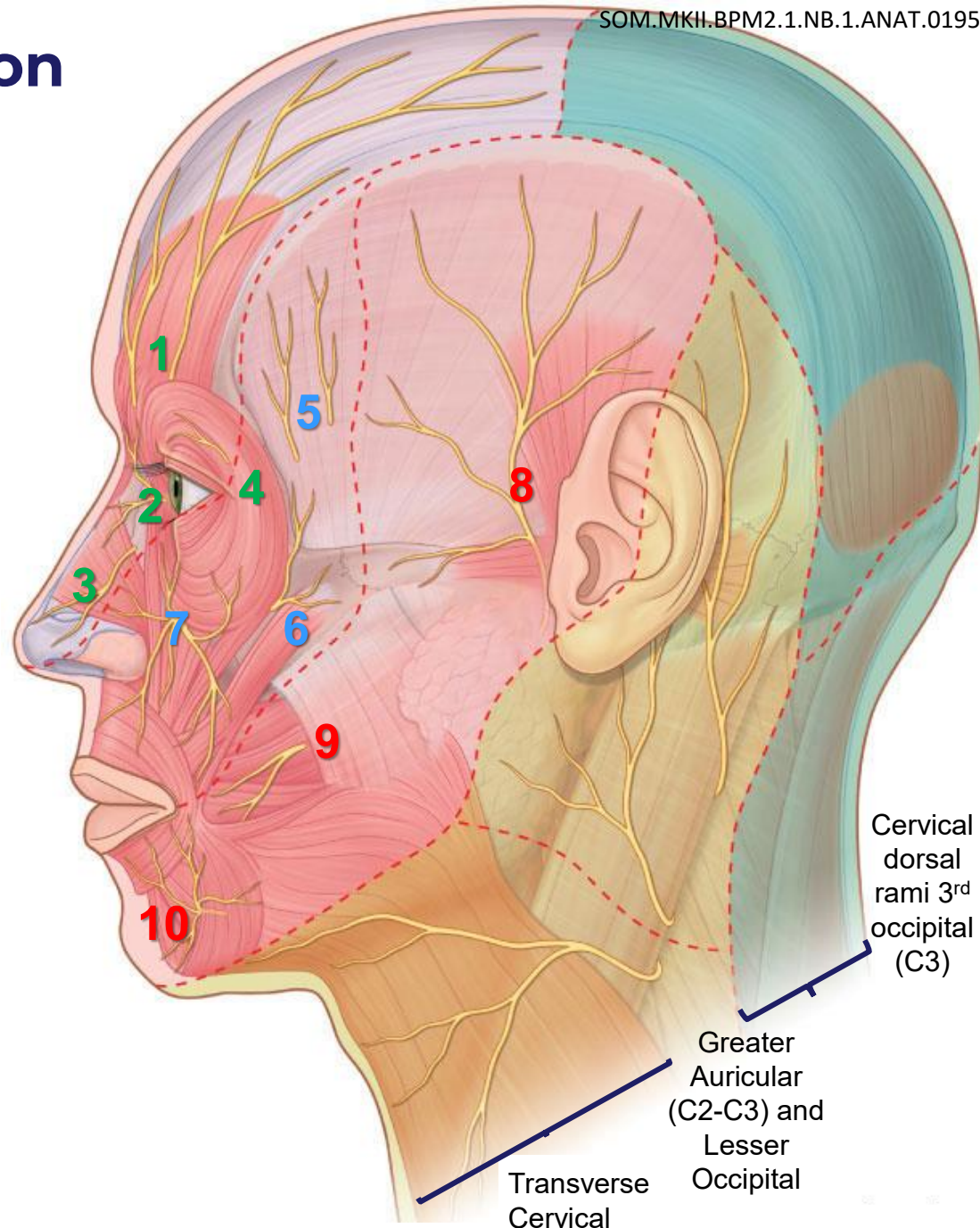
1. Supraorbital & Supratrochlear
2. Infratrochlear
3. External Nasal (from Anterior Ethmoid)
4. Lacrimal

Maxillary nerve (CN V₂)

5. Zygomaticotemporal
6. Zygomaticofacial
7. Infraorbital

Mandibular nerve (CN V₃)

8. Auriculotemporal
9. Buccal nerve
10. Mental



Clinical Correlate - Trigeminal Neuralgia

- Sudden attacks of severe pain (i.e. **tic douloureux**) in the face
 - due to reflexive spasmodic contraction of facial muscles in response to pain
- more commonly affects the mandibular (**V₃**) and maxillary (**V₂**) areas
 - causes include vasculature compressing the trigeminal nerve by the pons

Symptom characteristics include:

- typically unilateral
- duration: few seconds to two minutes
- frequency: regular, recurrent; sometimes several times a day, or for days to weeks

Treatment options:

- Medication management
- Surgical decompression
- Brain stimulation

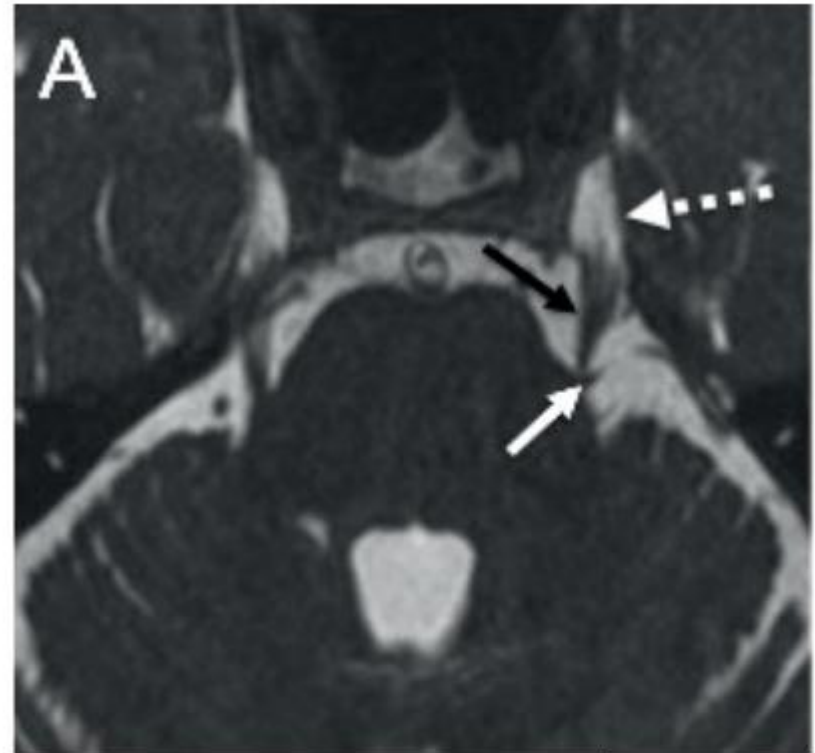


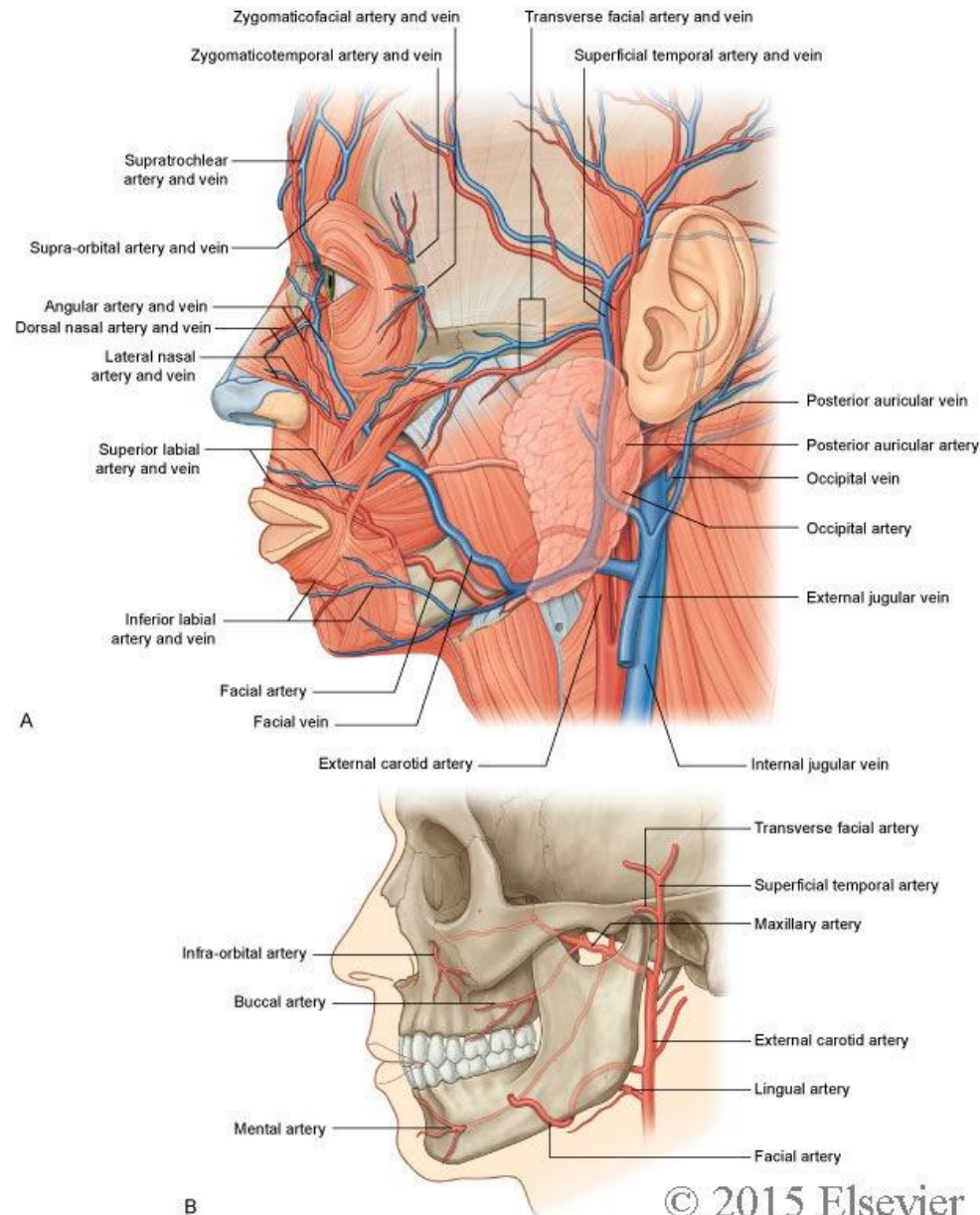
Figure 2. Neurovascular conflict causing TN.

(a) Axial CISS image shows a vascular loop (white arrow), the left superior cerebellar artery, mildly deforming the root entry zone of the cisternal segment of the left CN V (black arrow). Note the cisternal segment bilaterally extending anteriorly to enter the Meckel's cave (dashed arrow), which is filled with CSF.

from DOI: 10.17096/jiufd.27242

Face - Blood Supply

- External carotid artery
 - Facial artery
 - Angular artery
 - Superior labial branch
 - Inferior labial branch
 - Superficial temporal artery
 - Transverse facial artery
- Maxillary artery
 - Infra-orbital
 - Buccal
 - Mental
- Internal carotid artery
 - Ophthalmic artery
 - Supratrochlear
 - Supraorbital
 - Dorsal nasal



Face - Venous Drainage

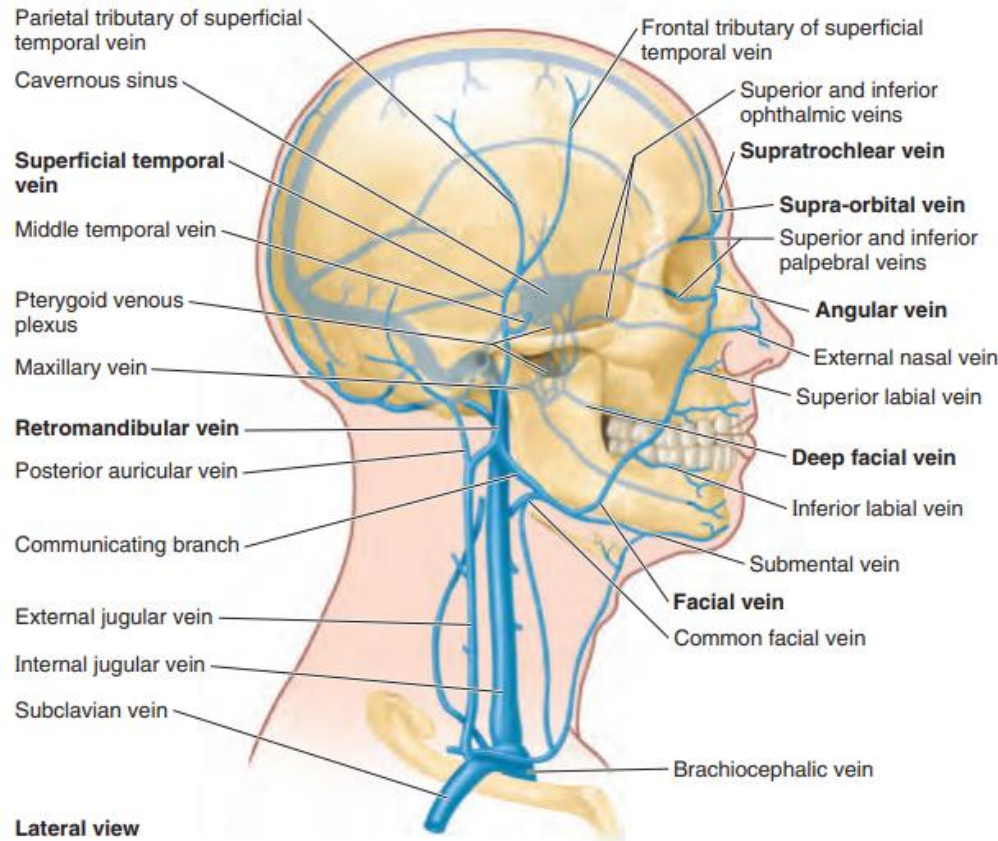


FIGURE 7.25. Veins of face and scalp.

All provide a possible route of entry for infections from face and scalp to deep intracranial venous structures

Facial vein

- Angular
- Supraorbital
- Supratrochlear

Transverse facial vein

- drains lateral cheek to superficial temporal vein

Facial vein communicates with **deeper veins**:

- ophthalmic veins
- infra-orbital vein
- pterygoid plexus
- cavernous sinus

Intracranial venous connections:

- Emissary veins connect extracranial and intracranial veins

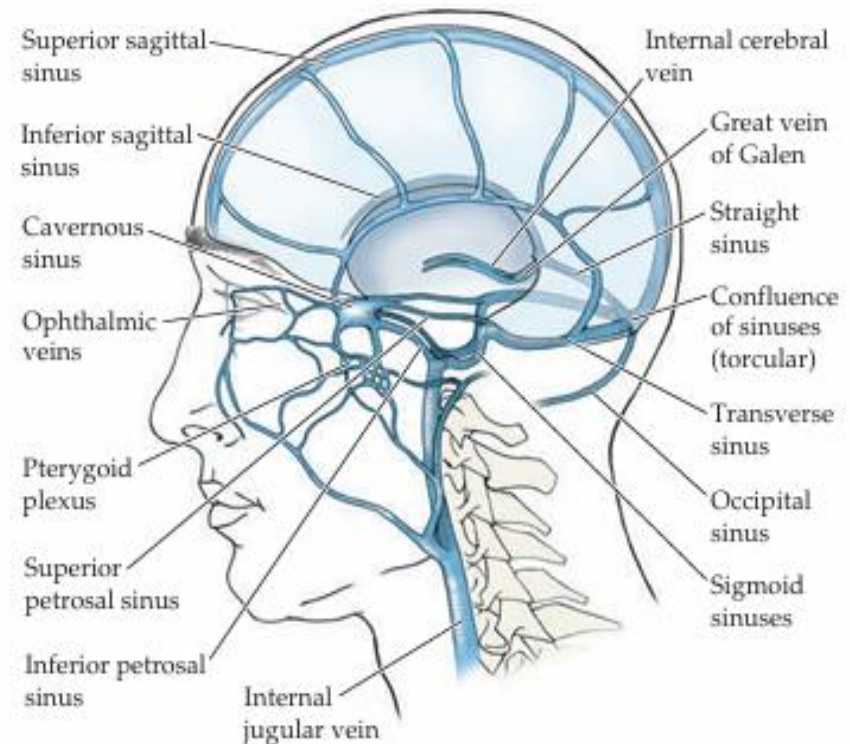
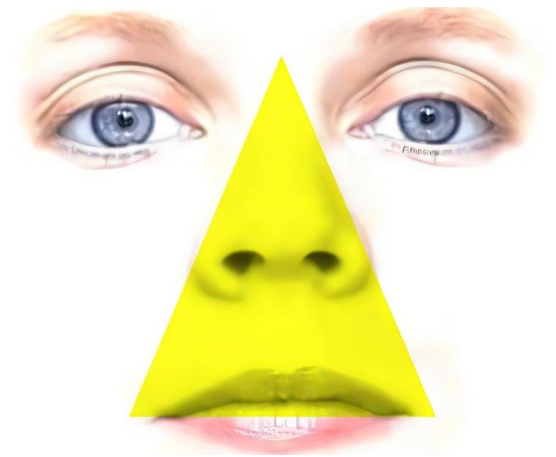
Danger Area of the Face

i.e. region between the upper lip and the lower eyelid.

Since veins of head are **valveless**, infections of the skin of the face can spread into the head by way of venous anastomoses of the facial vein with deep veins of the face

Infectious materials can be carried into the **cavernous sinus**, which is:

- adjacent to the deep surface of the brain
- surrounds the pituitary gland
- provides a passageway for several cranial nerves



Face – Lymphatic Drainage

Submental nodes

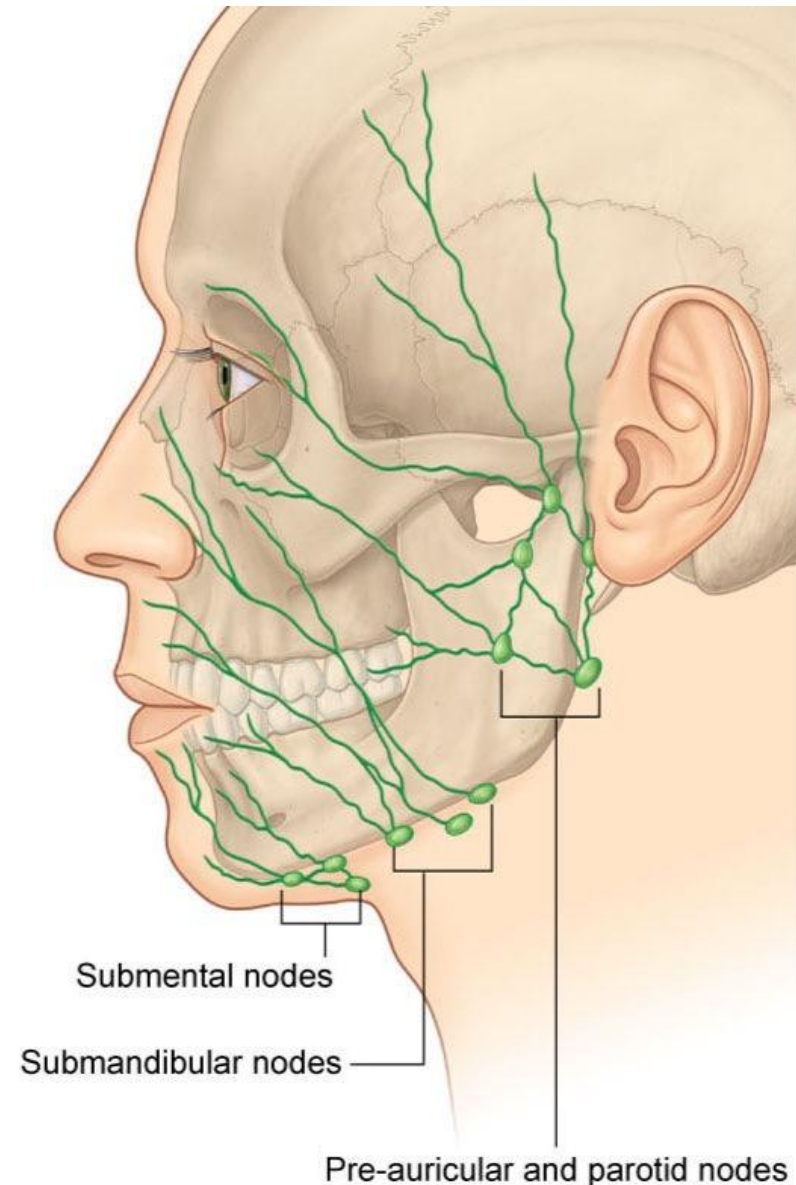
- medial lower lip and chin
- anterior portion of the tongue
- floor of the mouth

Submandibular nodes

- medial corner of the orbit
- most external nasal skin
- medial cheek
- upper lip, lateral lower lip

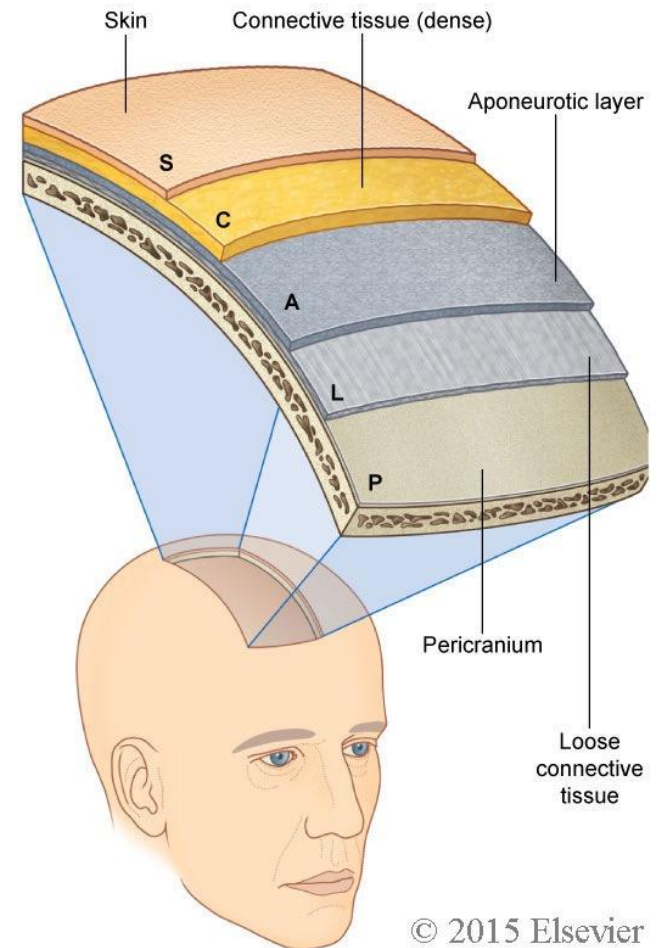
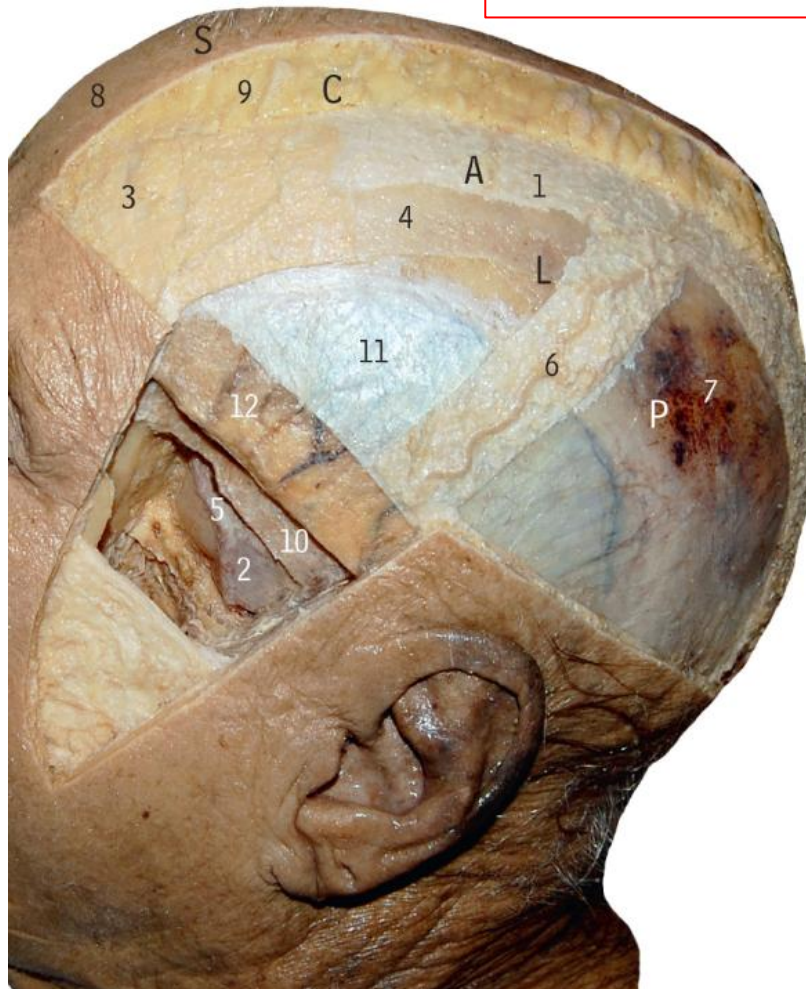
Pre-auricular and Parotid nodes

- eyelids
- root of nose
- lateral cheek



Scalp - Layers

Skin
Connective tissue (dense)
Aponeurotic layer
Loose connective tissue (areolar)
Pericranium



Scalp

Scalp Proper

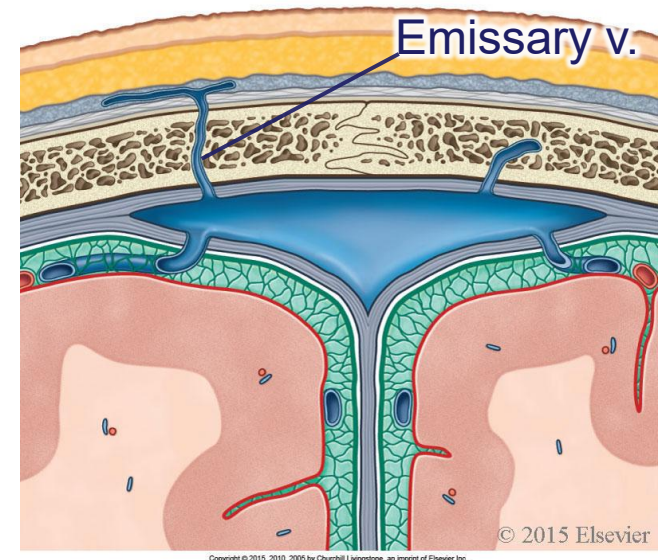
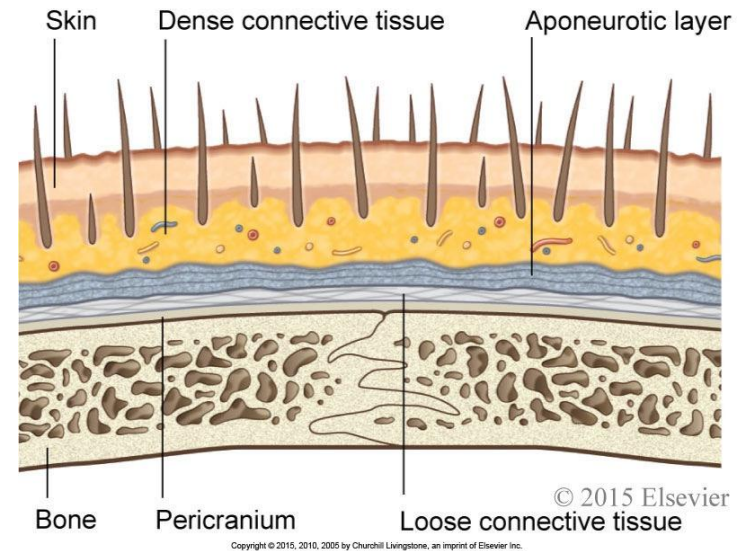
- Three layers tightly held together as a single unit
 - **S**kin: contains hair follicles
 - **C**onnective tissue (dense): joins skin to aponeurotic layer
 - **A**poneurotic layer: contains occipitofrontalis muscle

Loose connective tissue

- Facilitates movement of the scalp proper over the calvaria
 - facilitates scalp removal
- 'Danger zone'
 - infections may spread through this layer to emissary vein and cranium

Pericranium

- periosteum on the outer calvaria fuses at the suture lines



Danger Zone of the Scalp

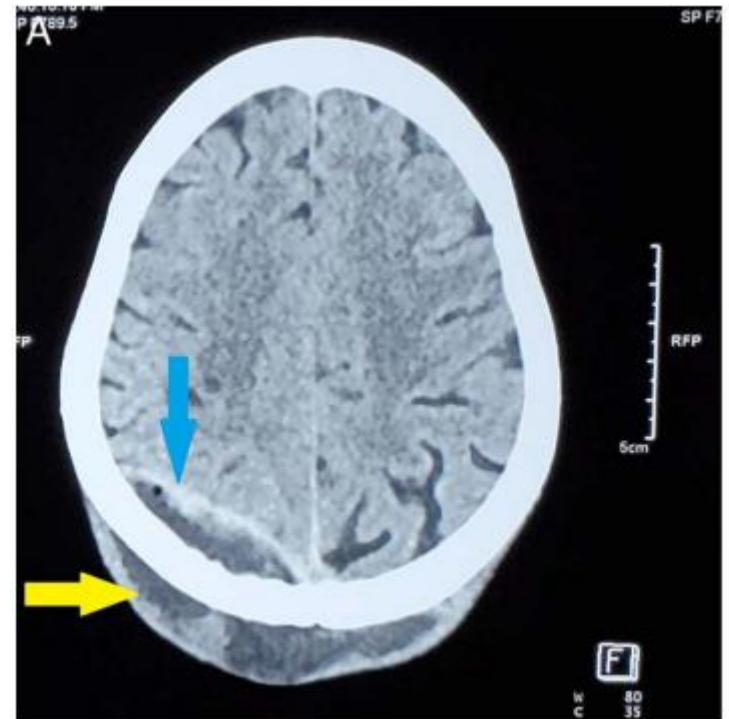
i.e. the loose areolar layer of connective tissue between the aponeurotic layer and the pericranium

Scalp has extremely rich blood supply from the external carotid arteries.

Lacerations of the scalp tend to bleed profusely because:

1. In the erect position venous pressure is extremely low
2. The vessels do not retract and close when lacerated since the dense connective tissue layer in which they are found holds them open

from DOI: 10.1097/MS9.0000000000001285



CT of subgaleal region
(yellow arrow)

Extracranial swellings

- Due to fluid accumulation or ruptured vessels
 - may be seen during vaginal deliveries
- Usually non-life threatening
 - no excessive pressure on the brain
- Tend to resolve with time
- Classified based on swelling location

Caput succedaneum

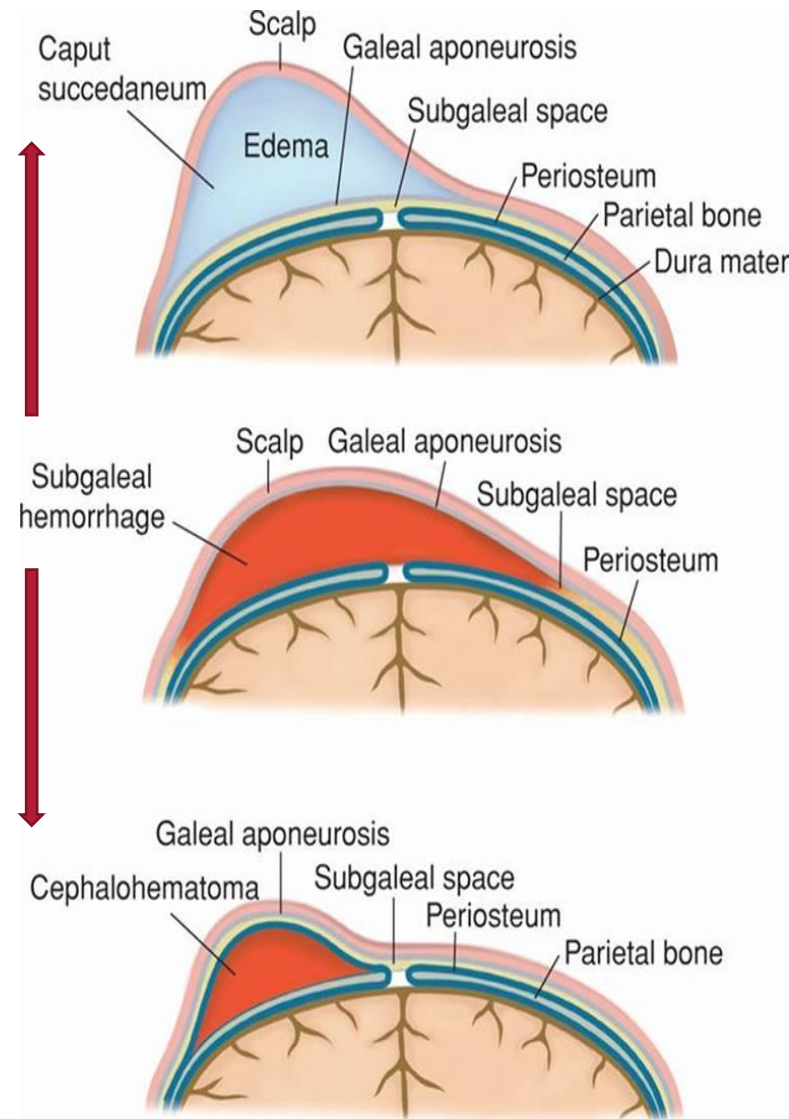
- Soft superficial swelling just under the skin
- Visible **from time of birth**

Subgaleal hemorrhage

- Hemorrhage between the aponeurotic layer and periosteum
- Develops **few hours after delivery**
- Can involve large amount of blood

Cephalohematoma

- Hemorrhage between the periosteum and bone
- Usually appears within **24-72 hours after delivery** and may take weeks to resolve
- **Does not cross suture lines** since periosteum fuses (confined to individual bones)



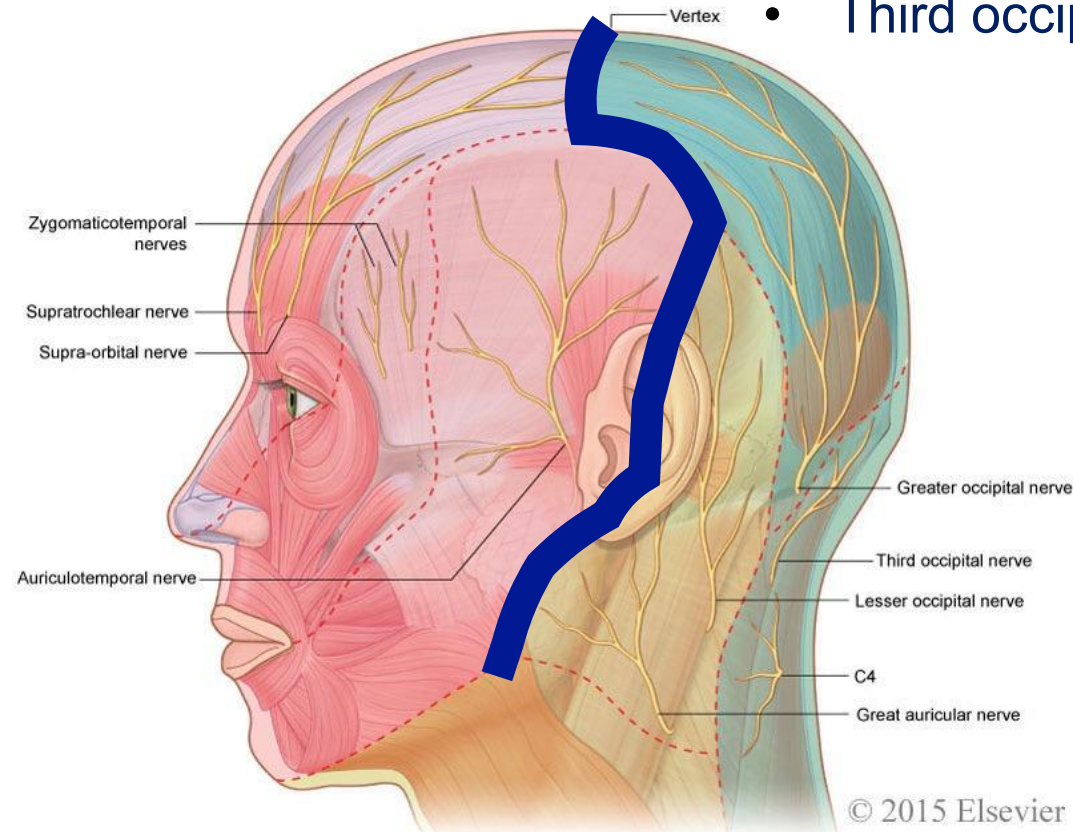
Scalp - Innervation

Anterior to the ears and the vertex =
CN V

- V_1 : Supratrochlear, Supraorbital
- V_2 : Zygomaticotemporal
- V_3 : Auriculotemporal

Posterior to the ears and the
vertex = branches from spinal
nerves (C2, C3)

- Great auricular
- Greater occipital
- Lesser occipital
- Third occipital



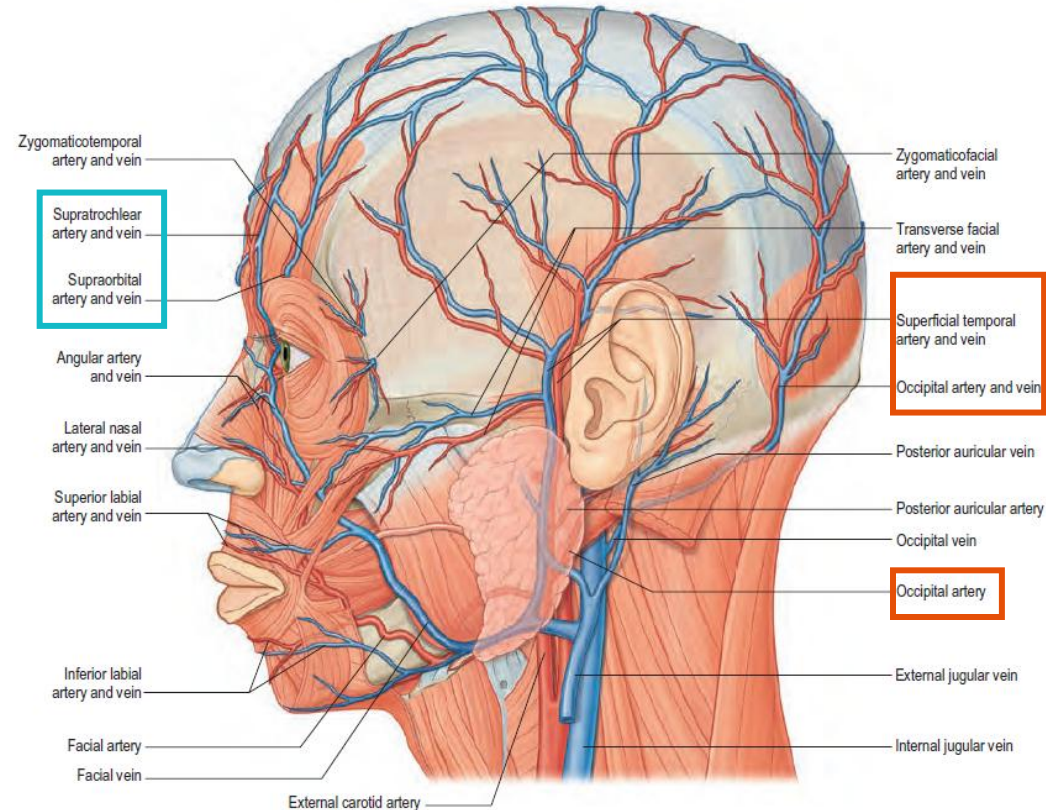
Scalp - Blood Supply

Branches of ECA

- Superficial temporal a.
- Posterior auricular a.
- Occipital a.

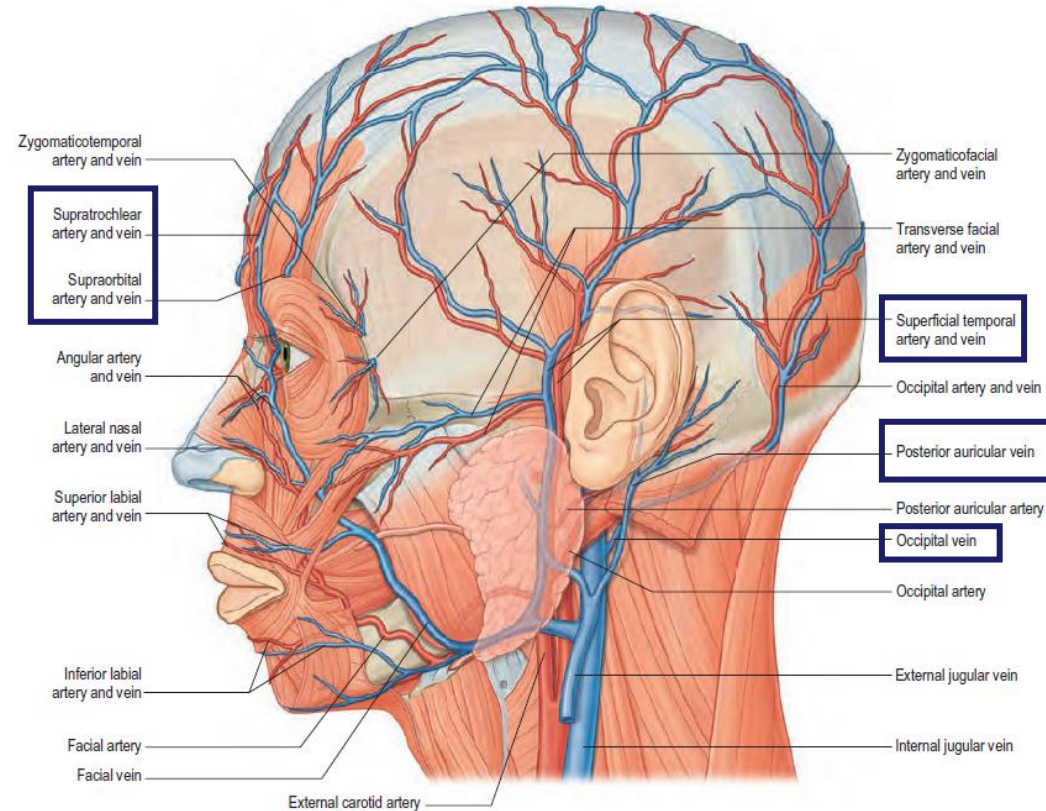
Ophthalmic (*branch of ICA*)

- Supratrochlear a.
- Supraorbital a.



Scalp - Venous Drainage

- Supratrochlear v.
- Supraorbital v.
- Posterior auricular v.
- Occipital v.
- **Superficial temporal v.**
 - drains lateral scalp
 - forms retromandibular vein



Scalp - Lymphatic Drainage

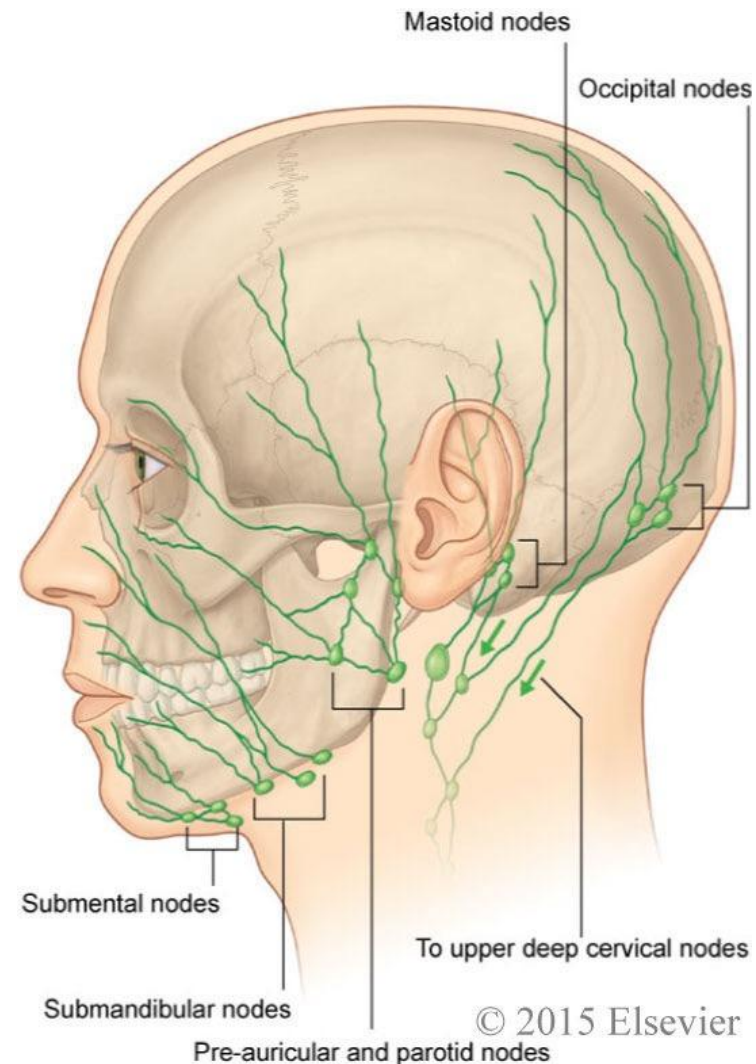
Generally follows arterial distribution

Occipital region

- occipital nodes
- upper deep cervical nodes

Upper part of the scalp

- Posterior to the vertex
 - mastoid nodes
 - upper deep cervical nodes
- Anterior to the vertex
 - pre-auricular
 - parotid nodes
- Forehead
 - Facial nodes to submandibular nodes following facial vessels.



© 2015 Elsevier

Copyright © 2015, 2010, 2005 by Churchill Livingstone, an imprint of Elsevier Inc.

fin

